SAT Math Mastery · Session 9

Right Triangles, Trigonometry and Circles

Time 3 hours: Part A in class (about 2 hours of solving), then Part B, the self-assessment, done independently (about 50 minutes), and a short error log.

Topics right-triangle trigonometry (sin, cos, tan and cofunctions) · special right triangles and the Pythagorean theorem · altitude to the hypotenuse · Laws of Sines and Cosines, area with sin · distance and applied right triangles · circle equations (completing the square) · arcs, chords, tangents and inscribed angles · radians and degrees

Name / Date _____ _____

PART A · IN CLASS — 36 PROBLEMS

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal). Grey boxes in the answer key show exactly what to type in Desmos, one expression per line; tap a grey dot to read an intersection, intercept or vertex. Calculator problems assume degree mode.

1. Medium · No calculator needed

If $\sin(2x)^\circ = \cos(x+15)^\circ$ and $0 < x < 45$, what is the value of x?

1. Sine of one angle equals cosine of its complement, so the two angles must add to 90°. Write $(2x) + (x+15) = 90$.

2. Combine like terms: $3x + 15 = 90$, so $3x = 75$.

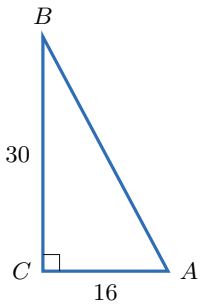
3. Divide by 3: $x = 25$.

4. Check: $\sin 50^\circ = \cos 40^\circ$ since $50 + 40 = 90$. The domain $0 < x < 45$ rules out the other branches (such as $x = 105$).

Answer: 25

2. Medium · No calculator needed

In right triangle ABC, the right angle is at C. If $AC = 16$ and $BC = 30$, what is $\tan A$?



(A) $\frac{16}{34}$

(B) $\frac{34}{16}$

(C) $\frac{8}{15}$

(D) $\frac{15}{8}$ ✓

1. $\tan A = \frac{\text{opposite}}{\text{adjacent}}$. Opposite angle A is $BC = 30$; adjacent is $AC = 16$.

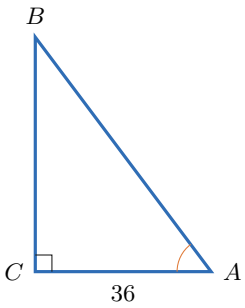
2. So $\tan A = \frac{30}{16} = \frac{15}{8}$ after dividing top and bottom by 2.

3. That is option (D). The option with $\frac{8}{15}$ flips the fraction, and the ones with 34 use the hypotenuse $\sqrt{16^2 + 30^2} = 34$.

Answer: (D)

3. HARD · NO CALCULATOR NEEDED

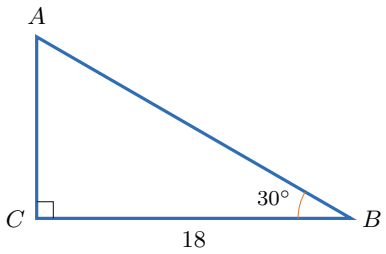
In right triangle ABC, the right angle is at C, and $\cos A = \dfrac{3}{5}$. If $AC = 36$, what is the length of AB?



1. $\cos A = \dfrac{\text{adjacent}}{\text{hypotenuse}} = \dfrac{AC}{AB}$, so $\dfrac{36}{AB} = \dfrac{3}{5}$.
2. Cross-multiply: $3 \cdot AB = 5 \cdot 36 = 180$.
3. Divide by 3: $AB = 60$. (Check: 36-48-60 is the 3-4-5 triangle times 12.)
Answer: 60

4. MEDIUM · NO CALCULATOR NEEDED

In triangle ABC, angle $C = 90°$, angle $B = 30°$, and $BC = 18$. What is the length of AC?



(A) 9

(B) $6\sqrt{3}$ ✓

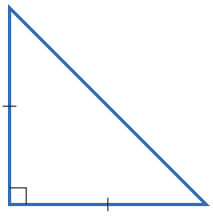
(C) $9\sqrt{3}$

(D) $18\sqrt{3}$

1. AC is opposite the $30°$ angle and $BC = 18$ is adjacent to it, so use $\tan 30° = \dfrac{AC}{18}$.
2. $\tan 30° = \dfrac{1}{\sqrt{3}}$, so $AC = \dfrac{18}{\sqrt{3}}$.
3. Rationalize: $\dfrac{18}{\sqrt{3}} \cdot \dfrac{\sqrt{3}}{\sqrt{3}} = \dfrac{18\sqrt{3}}{3} = 6\sqrt{3}$ (about 10.39). Option (B).
Answer: (B)

5. HARD · NO CALCULATOR NEEDED

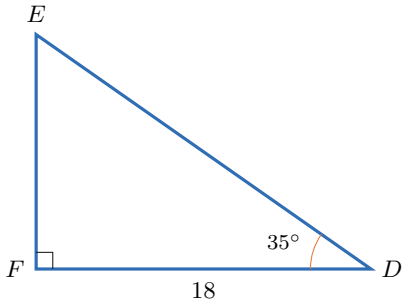
An isosceles right triangle has a perimeter of $40 + 20\sqrt{2}$. What is the length of the hypotenuse of the triangle?



1. Let each leg be L. The hypotenuse is $L\sqrt{2}$, so the perimeter is $2L + L\sqrt{2} = L(2 + \sqrt{2})$.
2. The given perimeter factors the same way: $40 + 20\sqrt{2} = 20(2 + \sqrt{2})$.
3. So $L = 20$ and the hypotenuse is $L\sqrt{2} = 20\sqrt{2}$.
Answer: $20\sqrt{2}$

6. HARD · CALCULATOR

In right triangle DEF, angle F is the right angle, angle D measures 35°, and side DF (adjacent to angle D) measures 18. Which of the following is closest to the length of side EF (opposite angle D)?



(A) 10.3

(B) 12.6 ✓

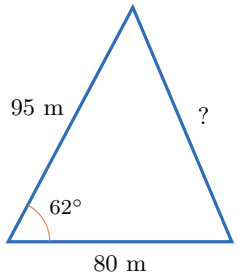
(C) 14.9

(D) 17.1

1. The two legs are involved (opposite EF, adjacent $DF = 18$), so use tangent: $\tan 35° = \dfrac{EF}{18}$.
2. Make sure the calculator is in degree mode: $\tan 35° \approx 0.7002$.
3. $EF = 18 \tan 35° \approx 18(0.7002) \approx 12.6$. Option (B).
Answer: (B)

7. HARD · CALCULATOR

A triangular plot of land has two sides measuring 80 meters and 95 meters, with an included angle of 62° between them. Which of the following is closest to the length, in meters, of the third side?



(A) 83.4

(B) 87.9

(C) 91.0 ✓

(D) 95.7

1. Two sides and the angle between them: use the Law of Cosines, $c^2 = 80^2 + 95^2 - 2(80)(95)\cos 62°$.

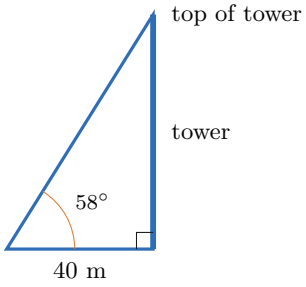
2. In degree mode $\cos 62° \approx 0.4695$, so $c^2 \approx 6400 + 9025 - 15200(0.4695) \approx 8289$.

3. $c \approx \sqrt{8289} \approx 91.0$. Option (C).

Answer: (C)

8. HARD · CALCULATOR

From a point on the ground 40 meters from the base of a communications tower, the angle of elevation to the top of the tower is 58°. Which of the following is closest to the height, in meters, of the tower?



(A) 33.9

(B) 47.1

(C) 64.0 ✓

(D) 75.5

1. The tower, the ground and the line of sight make a right triangle. The 40 meters is adjacent to the 58° angle and the height is opposite, so use $\tan 58° = \dfrac{h}{40}$.

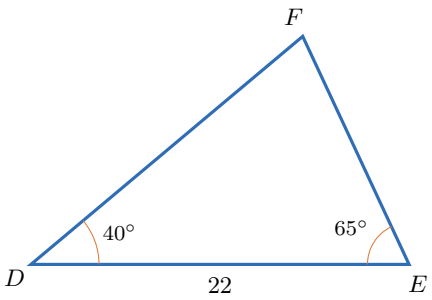
2. In degree mode $\tan 58° \approx 1.6003$.

3. $h = 40 \tan 58° \approx 40(1.6003) \approx 64.0$ meters. Option (C).

Answer: (C)

9. HARD · CALCULATOR

In triangle DEF, angle $D = 40°$, angle $E = 65°$, and side $DE = 22$. Which of the following is closest to the length of side EF (opposite angle D)?



(A) 12.6

(B) 14.6 ✓

(C) 16.9

(D) 19.4

1. The angles add to 180°, so $F = 180° - 40° - 65° = 75°$.

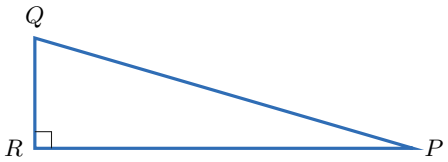
2. Law of Sines: $\dfrac{EF}{\sin 40°} = \dfrac{DE}{\sin F} = \dfrac{22}{\sin 75°}$ (side EF faces the 40° angle, side DE faces F).

3. $EF = 22 \cdot \dfrac{\sin 40°}{\sin 75°} \approx 22 \cdot \dfrac{0.6428}{0.9659} \approx 14.6$. Option (B).

Answer: (B)

10. HARD · NO CALCULATOR NEEDED

In right triangle PQR, angle R is the right angle. If $\sin(P) = \dfrac{7}{25}$, what is the value of $\cos(Q)$?



(A) $\dfrac{7}{25}$ ✓

(B) $\dfrac{24}{25}$

(C) $\dfrac{25}{7}$

(D) $\dfrac{25}{24}$

1. Angle R is the right angle, so $P + Q = 90°$: the two acute angles are complementary.

2. For complementary angles, the sine of one equals the cosine of the other: $\cos Q = \sin P$.

3. So $\cos Q = \dfrac{7}{25}$. Option (A). (Check with sides: $QR = 7$, $PQ = 25$, $\cos Q = \dfrac{QR}{PQ} = \dfrac{7}{25}$.)

Answer: (A)

11. HARD · NO CALCULATOR NEEDED

Maya walked from Noah's house 4 miles due north, then 3 miles due east, while Noah walked 2 miles due south and 5 miles due west. If Noah were to walk directly to Maya's location, how many hours would it take him, if Noah walks at a rate of 5 miles per hour?

1. Put Noah's house at the origin. Maya ends at $(3, 4)$ and Noah ends at $(-5, -2)$.

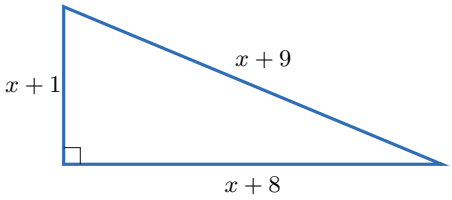
2. Distance: $\sqrt{(3-(-5))^2+(4-(-2))^2} = \sqrt{8^2+6^2} = \sqrt{100} = 10$ miles.

3. Time $= \dfrac{\text{distance}}{\text{rate}} = \dfrac{10}{5} = 2$ hours.

Answer: 2

12. HARD · NO CALCULATOR NEEDED

The legs of a right triangle have lengths $x+1$ and $x+8$, and the hypotenuse has length $x+9$. What is the area of the triangle?



1. Pythagorean theorem: $(x+1)^2 + (x+8)^2 = (x+9)^2$.

2. Expand: $2x^2 + 18x + 65 = x^2 + 18x + 81$, so $x^2 = 16$ and $x = 4$ (lengths are positive).

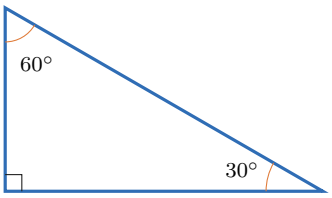
3. The legs are $x+1 = 5$ and $x+8 = 12$ (the hypotenuse is 13).

4. Area $= \dfrac{1}{2}(5)(12) = 30$.

Answer: 30

13. HARD · NO CALCULATOR NEEDED

A 30°-60°-90° triangle has a perimeter of $18 + 6\sqrt{3}$. What is the length of its hypotenuse?



(A) 6

(B) $6\sqrt{3}$

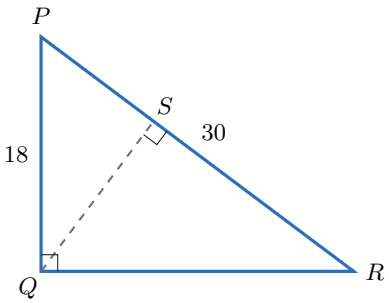
(C) 12 ✓

(D) $12\sqrt{3}$

1. In a 30°-60°-90° triangle the sides are s, $s\sqrt{3}$ and $2s$.
2. Perimeter: $s + s\sqrt{3} + 2s = 3s + s\sqrt{3} = s(3 + \sqrt{3})$. Since $18 + 6\sqrt{3} = 6(3 + \sqrt{3})$, we get $s = 6$.
3. Hypotenuse $= 2s = 12$. Option (C).
Answer: (C)

14. HARD · NO CALCULATOR NEEDED

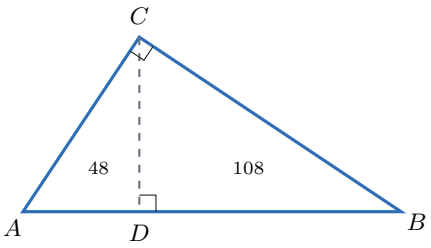
In right triangle PQR, the right angle is at Q, $PQ = 18$, and $PR = 30$. Point S lies on $\overline{PR}$ such that $\overline{QS} \perp \overline{PR}$. What is the value of $\dfrac{QR}{QS}$?



1. First $QR = \sqrt{30^2 - 18^2} = \sqrt{576} = 24$.
2. Area two ways: $\frac{1}{2}(PQ)(QR) = \frac{1}{2}(PR)(QS)$, so $18 \cdot 24 = 30 \cdot QS$ and $QS = 14.4$.
3. $\dfrac{QR}{QS} = \dfrac{24}{14.4} = \dfrac{5}{3}$ (or about 1.667).
Answer: $\frac{5}{3}$ **(or** 1.667)

15. HARD · NO CALCULATOR NEEDED

In triangle ABC, the angle at C is a right angle. Point D lies on $\overline{AB}$ so that $\overline{CD} \perp \overline{AB}$. The area of triangle ACD is 48 and the area of triangle CDB is 108. What is the length of $\overline{AB}$?



1. Triangles ACD and CDB share the height CD, so their areas are in the ratio of their bases: $\frac{AD}{DB} = \frac{48}{108} = \frac{4}{9}$.

2. Let $AD = 4k$ and $DB = 9k$. The altitude to the hypotenuse gives $CD^2 = AD \cdot DB = 36k^2$, so $CD = 6k$.

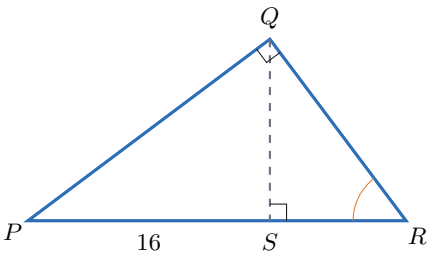
3. Area of ACD: $\frac{1}{2}(4k)(6k) = 12k^2 = 48$, so $k^2 = 4$ and $k = 2$.

4. $AB = AD + DB = 13k = 26$. (Check: $[CDB] = \frac{1}{2}(18)(12) = 108$.)

Answer: 26

16. HARD · NO CALCULATOR NEEDED

In triangle PQR, the angle at Q is a right angle. Point S lies on side $\overline{PR}$ so that $\overline{QS} \perp \overline{PR}$. If $\tan\angle QRP = \frac{4}{3}$ and $PS = 16$, what is the length of $\overline{SR}$?



(A) 12 (B) 15

(C) 25 **(D) 9 ✓**

1. $\tan\angle QRP = \frac{PQ}{QR} = \frac{4}{3}$, so the legs are $PQ = 4k$, $QR = 3k$ and the hypotenuse is $PR = 5k$.

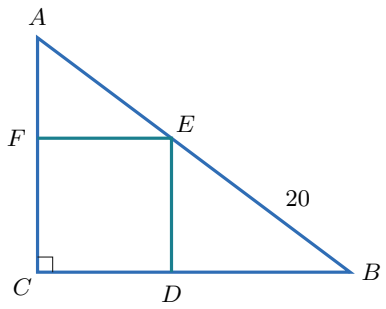
2. Leg rule: $PQ^2 = PS \cdot PR$, so $16k^2 = 16 \cdot 5k$, giving $k = 5$ and $PR = 25$.

3. $SR = PR - PS = 25 - 16 = 9$. Option (D). (Check: $QR^2 = SR \cdot PR$ gives $15^2 = 9 \cdot 25$.)

Answer: (D)

17. HARD · NO CALCULATOR NEEDED

In right triangle ABC, $\angle C = 90°$, $AB = 20$, and the area of the triangle is 96. Square $CDEF$ is inscribed in the triangle, with D on $\overline{BC}$, F on $\overline{AC}$, and E on $\overline{AB}$. What is the side length of the square?



(A) $\frac{48}{7}$ ✓ (B) $\frac{24}{7}$

(C) $\frac{48}{5}$ (D) $\frac{96}{7}$

1. Let the legs be a and b. The area gives $ab = 2 \cdot 96 = 192$ and Pythagoras gives $a^2 + b^2 = 20^2 = 400$.

2. $(a+b)^2 = a^2 + b^2 + 2ab = 400 + 384 = 784$, so $a + b = 28$.

3. The square's corner E lies on the hypotenuse, so $s\left(\frac{1}{a} + \frac{1}{b}\right) = 1$, which gives $s = \frac{ab}{a+b}$.

4. $s = \frac{192}{28} = \frac{48}{7}$. Option (A). (The legs are 12 and 16.)

Answer: (A)

18. HARD · NO CALCULATOR NEEDED

In right triangle PQR, $\angle R = 90°$ and $\sin P + \sin Q = \frac{\sqrt{6}}{2}$. What is the value of $\tan P + \tan Q$?

1. $Q = 90° - P$, so $\sin Q = \cos P$ and $\tan Q = \frac{1}{\tan P}$.

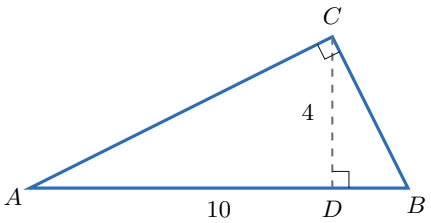
2. Square the given sum: $\sin^2 P + 2\sin P \cos P + \cos^2 P = \frac{6}{4}$, so $1 + 2\sin P \cos P = \frac{3}{2}$ and $\sin P \cos P = \frac{1}{4}$.

3. $\tan P + \tan Q = \frac{\sin P}{\cos P} + \frac{\cos P}{\sin P} = \frac{\sin^2 P + \cos^2 P}{\sin P \cos P} = \frac{1}{1/4} = 4$.

Answer: 4

19. HARD · NO CALCULATOR NEEDED

In right triangle ABC, the right angle is at C and $AB = 10$. Point D lies on $\overline{AB}$ with $\overline{CD} \perp \overline{AB}$ and $CD = 4$. If $AD > DB$, what is the value of $\cos A$?



(A) $\dfrac{\sqrt{5}}{5}$

(B) $\dfrac{4}{5}$

(C) $\dfrac{1}{2}$

(D) $\dfrac{2\sqrt{5}}{5}$ ✓

1. Let $AD = p$ and $DB = q$. Then $p + q = 10$ and $CD^2 = pq = 16$.

2. $p(10 - p) = 16$ gives $p^2 - 10p + 16 = 0$, so $(p - 2)(p - 8) = 0$. Since $AD > DB$, $AD = 8$ and $DB = 2$.

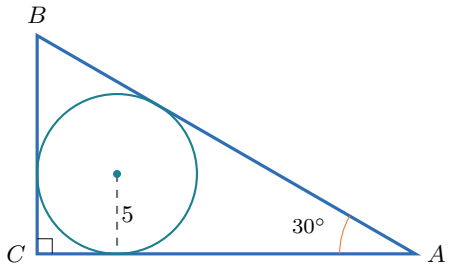
3. In triangle ACD: $AC = \sqrt{8^2 + 4^2} = \sqrt{80} = 4\sqrt{5}$.

4. $\cos A = \dfrac{AD}{AC} = \dfrac{8}{4\sqrt{5}} = \dfrac{2\sqrt{5}}{5}$. Option (D).

Answer: (D)

20. HARD · NO CALCULATOR NEEDED

In triangle ABC, angle C is a right angle and angle A measures 30°. The circle inscribed in the triangle has radius 5. What is the length of $\overline{AC}$?



(A) $15 + 5\sqrt{3}$ ✓

(B) $10 + 10\sqrt{3}$

(C) $10\sqrt{3} - 10$

(D) $5 + 5\sqrt{3}$

1. Let $BC = s$ (opposite 30°). Then $AC = s\sqrt{3}$ and $AB = 2s$.

2. For a right triangle, $r = \dfrac{BC + AC - AB}{2} = \dfrac{s + s\sqrt{3} - 2s}{2} = \dfrac{s(\sqrt{3} - 1)}{2} = 5$.

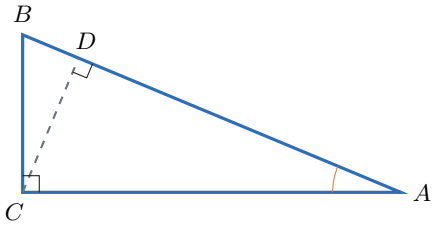
3. $s = \dfrac{10}{\sqrt{3} - 1} = \dfrac{10(\sqrt{3} + 1)}{2} = 5 + 5\sqrt{3}$.

4. $AC = s\sqrt{3} = 5\sqrt{3} + 15 = 15 + 5\sqrt{3}$ (about 23.66). Option (A).

Answer: (A)

21. HARD · NO CALCULATOR NEEDED

In right triangle ABC, the right angle is at C and $\tan A = \dfrac{5}{12}$. The area of the triangle is 120. Point D lies on $\overline{AB}$ such that $\overline{CD} \perp \overline{AB}$. What is the length of $\overline{AD}$?



(A) $\dfrac{120}{13}$

(B) 24

(C) $\dfrac{50}{13}$

(D) $\dfrac{288}{13}$ ✓

1. $\tan A = \dfrac{BC}{AC} = \dfrac{5}{12}$, so $BC = 5k$, $AC = 12k$ and $AB = 13k$.

2. Area: $\dfrac{1}{2}(12k)(5k) = 30k^2 = 120$, so $k = 2$ and $AC = 24$, $AB = 26$.

3. Leg rule: $AC^2 = AD \cdot AB$, so $AD = \dfrac{24^2}{26} = \dfrac{576}{26} = \dfrac{288}{13}$. Option (D).

Answer: (D)

22. HARD · NO CALCULATOR NEEDED

For $0 < x < 45$, the equation

$$\sin\left((5x - 20)^\circ\right) = \cos\left((x + 14)^\circ\right)$$

has more than one solution. What is the sum of all solutions x in this interval?

1. Rewrite the cosine as a sine: $\cos((x + 14)^\circ) = \sin((76 - x)^\circ)$.

2. Two angles with equal sines are either equal or supplementary (mod 360°). Equal: $5x - 20 = 76 - x$, so $6x = 96$ and $x = 16$.

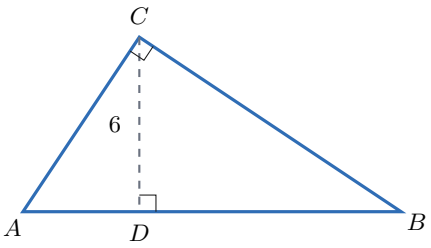
3. Supplementary: $5x - 20 = 180 - (76 - x) = 104 + x$, so $4x = 124$ and $x = 31$.

4. Both are in $0 < x < 45$ and other multiples of 360° give nothing in range. Sum: $16 + 31 = 47$.

Answer: 47

23. HARD · NO CALCULATOR NEEDED

In right triangle ABC, the angle at C is a right angle. Point D lies on $\overline{AB}$ so that $\overline{CD} \perp \overline{AB}$. The length of $\overline{CD}$ is 6, and DB is 5 units longer than AD. What is the value of $\sin B$?



(A) $\dfrac{2\sqrt{13}}{13}$ ✓

(B) $\dfrac{3\sqrt{13}}{13}$

(C) $\dfrac{4}{13}$

(D) $\dfrac{6}{13}$

1. Let $AD = t$ and $DB = t + 5$. The altitude rule gives $CD^2 = AD \cdot DB$: $36 = t(t + 5)$.

2. $t^2 + 5t - 36 = 0$ factors as $(t + 9)(t - 4) = 0$, so $t = 4$, $DB = 9$ and $AB = 13$.

3. $BC^2 = DB \cdot AB = 9 \cdot 13 = 117$, so in triangle CDB, $\sin B = \dfrac{CD}{BC} = \dfrac{6}{3\sqrt{13}} = \dfrac{2\sqrt{13}}{13}$.

4. Option (A). (Check: $AC^2 = 4 \cdot 13 = 52$ and $\dfrac{AC}{AB} = \dfrac{2\sqrt{13}}{13}$.)

Answer: (A)

24. HARD · NO CALCULATOR NEEDED

In triangle PQR, the measure of angle P is $30°$ and $PR = 14$. The length $QR = a$ is a positive integer with $a \leq 20$. For how many values of a does the given information determine exactly one triangle, up to congruence?

1. This is the side-side-angle situation: $P = 30°$, adjacent side $PR = 14$, opposite side $QR = a$. The height from R to line PQ is $14 \sin 30° = 7$.

2. $a < 7$: QR cannot reach, no triangle. $a = 7$: it just touches, exactly one (right) triangle.

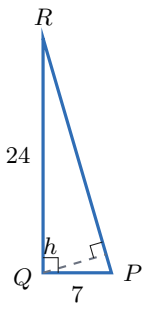
3. $7 < a < 14$: two different triangles. $a = 14$ or more: only one position for Q works.

4. One triangle for $a = 7$ and $a = 14, 15, \ldots, 20$: that is $1 + 7 = 8$ values.

Answer: 8

25. HARD · NO CALCULATOR NEEDED

In right triangle PQR, the right angle is at Q, $PQ = 7$ and $QR = 24$. The altitude from Q to hypotenuse $\overline{PR}$ has length h. What is the value of h? (Enter as a decimal.)



1. Hypotenuse: $PR = \sqrt{7^2 + 24^2} = \sqrt{625} = 25$.

2. Area two ways: $\frac{1}{2}(7)(24) = \frac{1}{2}(25)h$, so $25h = 168$.

3. $h = \frac{168}{25} = 6.72$.

Answer: 6.72

26. HARD · NO CALCULATOR NEEDED

In a right triangle, angle θ is acute and $\sin\theta = 0.6$. What is the value of $\cos(90° - \theta) + \tan\theta$?

(A) 1.35 ✓ (B) 1.6

(C) 0.75 (D) 1.2

1. $\cos(90° - \theta) = \sin\theta = 0.6$.

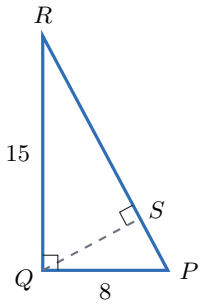
2. $\sin\theta = \frac{3}{5}$, so the sides are $3, 4, 5$ and $\cos\theta = \frac{4}{5}$, $\tan\theta = \frac{3}{4} = 0.75$.

3. Sum: $0.6 + 0.75 = 1.35$. Option (A).

Answer: (A)

27. HARD · NO CALCULATOR NEEDED

In right triangle PQR, the right angle is at Q, $PQ = 8$ and $QR = 15$. Point S is on $\overline{PR}$ such that $\overline{QS}$ is perpendicular to $\overline{PR}$. What is the length of $\overline{QS}$? (Enter as a fraction.)



1. $PR = \sqrt{8^2 + 15^2} = \sqrt{289} = 17$.

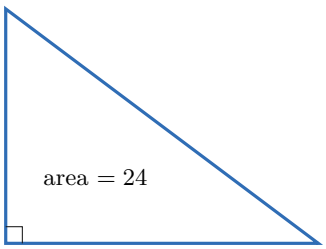
2. Area two ways: $\frac{1}{2}(8)(15) = \frac{1}{2}(17)(QS)$, so $17 \cdot QS = 120$.

3. $QS = \frac{120}{17}$.

Answer: $\frac{120}{17}$

28. HARD · NO CALCULATOR NEEDED

In a right triangle, the sine of one acute angle is $\frac{3}{5}$ and the area of the triangle is 24. What is the length of the hypotenuse?



1. $\sin = \frac{3}{5}$ means the sides are $3k$, $4k$ and $5k$ (a 3-4-5 triangle).

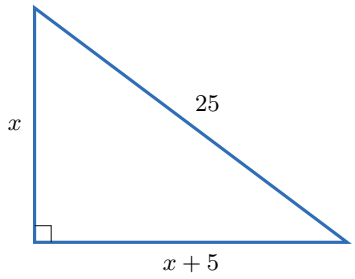
2. Area $= \frac{1}{2}(3k)(4k) = 6k^2 = 24$, so $k^2 = 4$ and $k = 2$.

3. Hypotenuse $= 5k = 10$.

Answer: 10

29. HARD · DESMOS

The legs of a right triangle have lengths x and $x+5$, and the hypotenuse is 25. What is the area of the triangle?



1. Pythagorean theorem: $x^2+(x+5)^2=25^2$. Type `y=x^2+(x+5)^2` and `y=625` in Desmos.
2. Tap the intersection with a positive x: (15, 625). So $x=15$ and the legs are 15 and 20 (the other intersection has $x=-20$, impossible for a length).
3. Area $=\frac{1}{2}(15)(20)=150$.

Answer: 150

30. HARD · NO CALCULATOR NEEDED

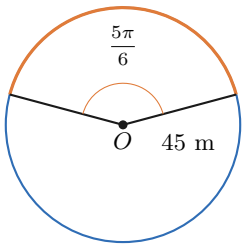
In the xy-plane, the graph of $x^2+y^2+4x-10y+k=0$ is a circle with radius 3. What is the value of k?

1. Complete the square: $x^2+4x=(x+2)^2-4$ and $y^2-10y=(y-5)^2-25$.
2. So $(x+2)^2+(y-5)^2=29-k$.
3. The radius is 3, so $r^2=9=29-k$ and $k=20$.

Answer: 20

31. HARD · NO CALCULATOR NEEDED

A circular plaza has a radius of 45 meters. A decorative walking path forms an arc along the plaza's boundary that subtends a central angle of $\frac{5\pi}{6}$ radians. What is the length of the arc, in meters?



(A) $\frac{25\pi}{2}$

(B) $\frac{75\pi}{2}$ ✓

(C) 45π

(D) 150π

1. Arc length $s = r\theta$ when θ is in radians.

2. $s = 45 \cdot \frac{5\pi}{6} = \frac{225\pi}{6} = \frac{75\pi}{2}$ meters. Option (B).

Answer: (B)

32. HARD · NO CALCULATOR NEEDED

A circle in the xy-plane has the equation $x^2 + y^2 - 8x + 2y - 19 = 0$. What is the diameter of the circle?

1. Complete the square: $x^2 - 8x = (x-4)^2 - 16$ and $y^2 + 2y = (y+1)^2 - 1$.

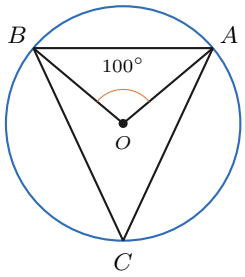
2. $(x-4)^2 + (y+1)^2 = 16 + 1 + 19 = 36$, so the radius is 6.

3. Diameter $= 2r = 12$.

Answer: 12

33. HARD · NO CALCULATOR NEEDED

In the circle above, chord AB subtends a central angle of $100°$ at center O. Point C lies on the major arc. What is the measure of angle ACB, in degrees?



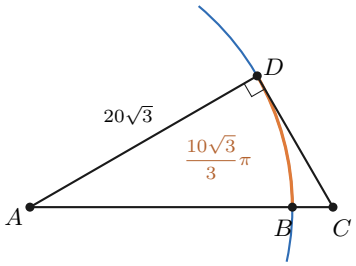
1. Inscribed Angle Theorem: an inscribed angle is half the central angle on the same arc.

2. Angle ACB looks at the minor arc AB (the one cut off by the $100°$ central angle), so $\angle ACB = \dfrac{100°}{2} = 50°$.

Answer: 50

34. HARD · NO CALCULATOR NEEDED

A circle has center A and radius $20\sqrt{3}$. Segment CD is tangent to the circle at point D, so triangle ACD has a right angle at D. Point B lies on the circle along ray AC, and minor arc BD has length $\dfrac{10\sqrt{3}}{3}\pi$. What is the length of CD?



1. Circumference: $2\pi(20\sqrt{3}) = 40\sqrt{3}\,\pi$. The arc is $\dfrac{10\sqrt{3}\,\pi/3}{40\sqrt{3}\,\pi} = \dfrac{1}{12}$ of it.

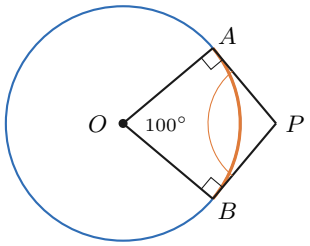
2. So $\angle DAC = \dfrac{360°}{12} = 30°$.

3. The tangent is perpendicular to the radius, so triangle ACD is a $30°$-$60°$-$90°$ triangle. CD is opposite $30°$ and $AD = 20\sqrt{3}$ is the longer leg, so $CD = \dfrac{20\sqrt{3}}{\sqrt{3}} = 20$.

Answer: 20

35. HARD · NO CALCULATOR NEEDED

In a figure, PA and PB are tangent to a circle centered at O, at points A and B respectively, so $OA \perp PA$ and $OB \perp PB$. If $\angle APB = 100°$ and the circle has a circumference of 36π, what is the length of minor arc AB?



(A) 6π

(B) 8π ✓

(C) 9π

(D) 10π

1. Circumference 36π means radius 18, but we only need the central angle.

2. Quadrilateral $PAOB$ has angles $90°$ (at A), $90°$ (at B) and $100°$ (at P), which add to $280°$, so $\angle AOB = 360° - 280° = 80°$.

3. Arc $= \frac{80}{360} \cdot 36\pi = 8\pi$. Option (B).

Answer: (B)

36. HARD · NO CALCULATOR NEEDED

A circle has equation $(x+4)^2 + (y-b)^2 = 100$, where b is a constant. The points $(4, 13)$ and $(-10, -1)$ both lie on the circle. What is the value of b?

1. The center is $(-4, b)$ and $r^2 = 100$. Plug in $(4, 13)$: $8^2 + (13-b)^2 = 100$, so $(13-b)^2 = 36$ and $b = 7$ or $b = 19$.

2. Plug in $(-10, -1)$: $(-6)^2 + (-1-b)^2 = 100$, so $(b+1)^2 = 64$ and $b = 7$ or $b = -9$.

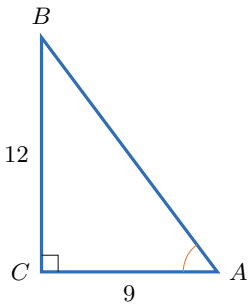
3. The value in both lists is $b = 7$.

Answer: 7

PART B · SELF-ASSESSMENT — 21 PROBLEMS

37. MEDIUM · NO CALCULATOR NEEDED

In right triangle ABC, angle C is the right angle, $AC = 9$, and $BC = 12$. What is $\sin(A)$?



(A) $\frac{4}{5}$ ✓

(B) $\frac{3}{4}$

(C) $\frac{4}{3}$

(D) $\frac{3}{5}$

1. Find the hypotenuse: $AB = \sqrt{9^2 + 12^2} = \sqrt{225} = 15$.

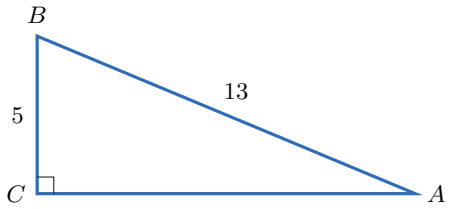
2. $\sin A = \frac{\text{opposite}}{\text{hypotenuse}}$, and the side opposite A is $BC = 12$.

3. $\sin A = \frac{12}{15} = \frac{4}{5}$. Option (A). The option $\frac{3}{5}$ is $\cos A$ (adjacent over hypotenuse).

Answer: (A)

38. MEDIUM · NO CALCULATOR NEEDED

In right triangle ABC, angle C is the right angle, $AB = 13$ (the hypotenuse), and $BC = 5$. What is $\cos(A)$?



(A) $\frac{5}{12}$

(B) $\frac{12}{5}$

(C) $\frac{5}{13}$

(D) $\frac{12}{13}$ ✓

1. Find the missing leg: $AC = \sqrt{13^2 - 5^2} = \sqrt{144} = 12$.

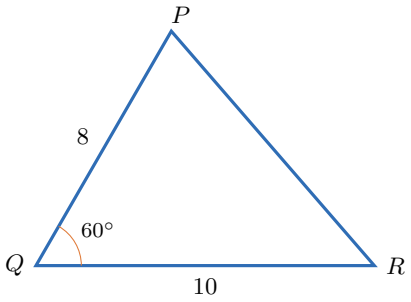
2. $\cos A = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{AC}{AB}$.

3. $\cos A = \frac{12}{13}$. Option (D). The option $\frac{5}{13}$ is $\sin A$.

Answer: (D)

39. Medium · No calculator needed

In $\triangle PQR$, $PQ = 8$, $QR = 10$, and $m\angle Q = 60°$. What is the area of $\triangle PQR$?



(A) 20 (B) $20\sqrt{2}$

(C) $20\sqrt{3}$ ✓ (D) $40\sqrt{3}$

1. Area of a triangle from two sides and the angle between them: $\frac{1}{2} \cdot PQ \cdot QR \cdot \sin Q$.

2. $\frac{1}{2}(8)(10)\sin 60° = 40 \cdot \frac{\sqrt{3}}{2}$.

3. $= 20\sqrt{3}$. Option (C).

Answer: (C)

40. Medium · No calculator needed

What is the distance between the points $(-3, -9)$ and $(9, -4)$?

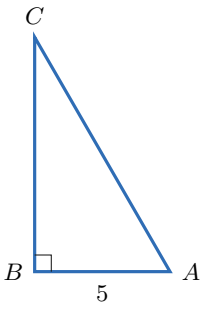
1. Subtract coordinates: $\Delta x = 9 - (-3) = 12$ and $\Delta y = -4 - (-9) = 5$.

2. Distance $= \sqrt{12^2 + 5^2} = \sqrt{169} = 13$.

Answer: 13

41. Medium · No calculator needed

In triangle ABC, angle B is a right angle and the measure of angle A is twice the measure of angle C. If $AB = 5$, what is the length of BC?



(A) 5

(B) $5\sqrt{2}$

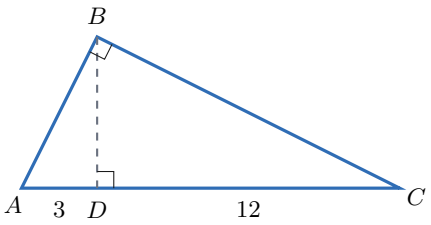
(C) $5\sqrt{3}$ ✓

(D) 10

1. The acute angles add to $90°$ and $A = 2C$, so $3C = 90°$, giving $C = 30°$ and $A = 60°$.
2. $AB = 5$ is opposite the $30°$ angle, so it is the short leg. BC is opposite the $60°$ angle.
3. Short leg times $\sqrt{3}$: $BC = 5\sqrt{3} \approx 8.66$. Option (C).
Answer: (C)

42. Medium · No calculator needed

In right triangle ABC, the right angle is at B, and BD is the altitude drawn to hypotenuse AC (with D on AC). If $AD = 3$ and $DC = 12$, what is the length of BD?



(A) 4

(B) 6 ✓

(C) 7.5

(D) 9

1. The altitude to the hypotenuse is the geometric mean of the two pieces: $BD^2 = AD \cdot DC$.
2. $BD^2 = 3 \cdot 12 = 36$.
3. $BD = 6$. Option (B). (Check: $AB^2 = 3^2 + 6^2 = 45$, $BC^2 = 12^2 + 6^2 = 180$, and $45 + 180 = 225 = 15^2$.)
Answer: (B)

43. Medium · No calculator needed

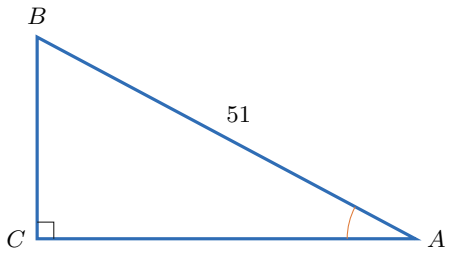
$$\sin(x^\circ) = \cos\left((2x+9)^\circ\right)$$

In the equation above, x is between 0 and 30. What is the value of x?

1. Sine of an angle equals cosine of its complement, so $x + (2x + 9) = 90$.
2. $3x + 9 = 90$, so $3x = 81$.
3. $x = 27$. Check: $2(27) + 9 = 63$ and $27 + 63 = 90$, and 27 is between 0 and 30.
Answer: 27

44. Medium · No calculator needed

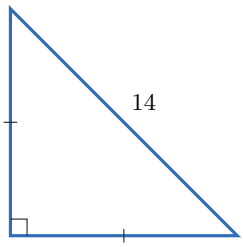
In right triangle ABC, the right angle is at C, $\tan A = \dfrac{8}{15}$, and $AB = 51$. What is the perimeter of triangle ABC?



1. $\tan A = \dfrac{BC}{AC} = \dfrac{8}{15}$, so the sides are $8k$, $15k$ and $\sqrt{8^2 + 15^2}\,k = 17k$.
2. The hypotenuse is $17k = 51$, so $k = 3$.
3. Legs: $8k = 24$ and $15k = 45$. Perimeter $= 24 + 45 + 51 = 120$.
Answer: 120

45. Medium · No calculator needed

An isosceles right triangle has a hypotenuse of length 14. What is the perimeter of the triangle?



(A) $14 + 7\sqrt{3}$

(B) 28

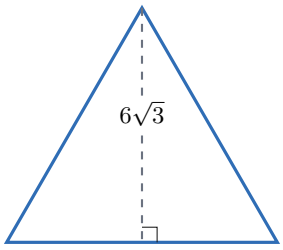
(C) $14 + 7\sqrt{2}$

(D) $14 + 14\sqrt{2}$ ✓

1. Each leg is hypotenuse over $\sqrt{2}$: $\dfrac{14}{\sqrt{2}} = 7\sqrt{2}$.
2. Perimeter $= 14 + 7\sqrt{2} + 7\sqrt{2} = 14 + 14\sqrt{2}$ (about 33.8). Option (D).
Answer: (D)

46. Medium · No calculator needed

The height of an equilateral triangle is $6\sqrt{3}$. What is the perimeter of the triangle?



1. The height of an equilateral triangle with side s is $\frac{s\sqrt{3}}{2}$, so $\frac{s\sqrt{3}}{2} = 6\sqrt{3}$.

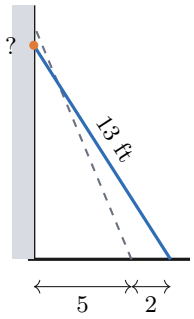
2. Divide both sides by $\sqrt{3}$: $\frac{s}{2} = 6$, so $s = 12$.

3. Perimeter $= 3s = 36$.

Answer: 36

47. Medium · Calculator

A 13-foot ladder leans against a vertical wall. Its base is 5 feet from the wall. The base is then pulled 2 feet farther from the wall, so the ladder is 7 feet from the wall. To the nearest tenth of a foot, how high up the wall does the ladder reach now?



1. The ladder is the hypotenuse (13 ft) and the base distance is a leg (7 ft): height $= \sqrt{13^2 - 7^2}$.

2. $13^2 - 7^2 = 169 - 49 = 120$.

3. $\sqrt{120} \approx 10.954$, which is 11.0 to the nearest tenth.

Answer: 11.0

48. Medium · No calculator needed

What is the value of $\cos\dfrac{5\pi}{3}$?

(A) $-\dfrac{\sqrt{3}}{2}$ (B) $-\dfrac{1}{2}$

(C) $\dfrac{\sqrt{3}}{2}$ **(D) $\dfrac{1}{2}$ ✓**

1. Subtract from a full turn: $\dfrac{5\pi}{3} = 2\pi - \dfrac{\pi}{3}$, a fourth-quadrant angle with reference angle $\dfrac{\pi}{3}$.

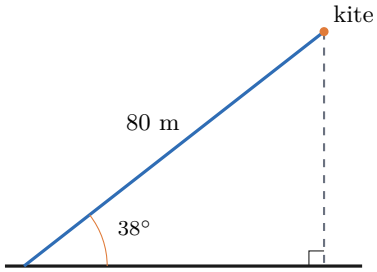
2. Cosine is positive in the fourth quadrant and $\cos\dfrac{\pi}{3} = \dfrac{1}{2}$.

3. So $\cos\dfrac{5\pi}{3} = \dfrac{1}{2}$. Option (D).

Answer: (D)

49. Medium · Calculator

A kite is flying at the end of a straight string that is 80 meters long. The string makes an angle of 38° with level ground. To the nearest meter, how high above the ground is the kite?



1. The string is the hypotenuse and the height is opposite the 38° angle, so use sine: $\sin 38° = \dfrac{h}{80}$.

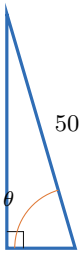
2. In degree mode $\sin 38° \approx 0.6157$.

3. $h = 80\sin 38° \approx 80(0.6157) \approx 49.25$, which is 49 meters to the nearest meter.

Answer: 49

50. Medium · No calculator needed

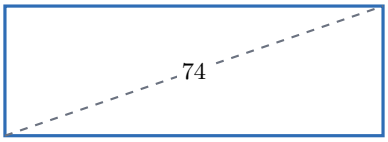
In a right triangle, the hypotenuse is 50 and the cosine of an acute angle θ is $\frac{7}{25}$. What is the area of the triangle?



1. $\cos\theta = \frac{7}{25}$ means adjacent over hypotenuse, so the adjacent leg is $50 \cdot \frac{7}{25} = 14$.
2. Other leg: $\sqrt{50^2 - 14^2} = \sqrt{2304} = 48$.
3. Area $= \frac{1}{2}(14)(48) = 336$.
Answer: 336

51. Medium · No calculator needed

The diagonal of a rectangle is 74 and the ratio of its two side lengths is $12 : 35$. What is the area of the rectangle?



1. Call the sides $12k$ and $35k$. The diagonal is $\sqrt{(12k)^2 + (35k)^2} = \sqrt{1369}\,k = 37k$.
2. $37k = 74$, so $k = 2$ and the sides are 24 and 70.
3. Area $= 24 \cdot 70 = 1680$.
Answer: 1680

52. Medium · No calculator needed

A circle in the xy-plane is defined by the equation $x^2 + y^2 - 6x + 8y = 0$. What is the radius of the circle?

(A) 3 (B) 4

(C) 5 ✓ (D) 25

1. Complete the square: $x^2 - 6x = (x-3)^2 - 9$ and $y^2 + 8y = (y+4)^2 - 16$.
2. So $(x-3)^2 + (y+4)^2 = 9 + 16 = 25$.
3. $r^2 = 25$, so $r = 5$. Option (C). (The option 25 is r^2.)
Answer: (C)

53. Medium · No calculator needed

A circle in the xy-plane is defined by the equation $x^2 + y^2 + 10x - 4y + 13 = 0$. What is the radius of the circle?

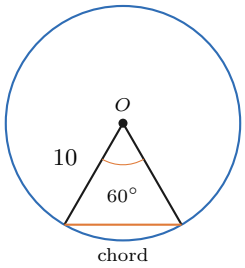
(A) 4 ✓ (B) 8

(C) 13 (D) 16

1. Complete the square: $x^2 + 10x = (x+5)^2 - 25$ and $y^2 - 4y = (y-2)^2 - 4$.
2. $(x+5)^2 + (y-2)^2 = 25 + 4 - 13 = 16$.
3. $r = \sqrt{16} = 4$. Option (A).
Answer: (A)

54. Medium · No calculator needed

In a circle with radius 10, a chord subtends a central angle of 60°. What is the length of the chord?



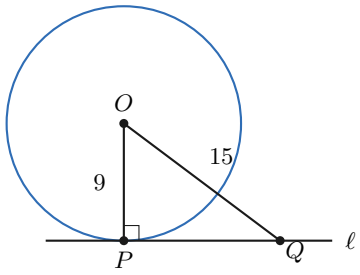
(A) 5 (B) $5\sqrt{3}$

(C) 10 ✓ (D) $10\sqrt{3}$

1. The two radii and the chord form an isosceles triangle with a 60° vertex angle, so the base angles are $\dfrac{180° - 60°}{2} = 60°$ each.
2. All three angles are 60°: the triangle is equilateral, so the chord equals the radius, 10. Option (C).
Answer: (C)

55. Medium · No calculator needed

In the figure, line ℓ is tangent to the circle with center O at point P. If $OP = 9$ and $OQ = 15$ for a point Q on line ℓ, what is the length of PQ?



(A) 6

(B) 12 ✓

(C) $\sqrt{306}$

(D) 18

1. A tangent is perpendicular to the radius at the point of tangency, so triangle OPQ has a right angle at P.
2. $PQ = \sqrt{OQ^2 - OP^2} = \sqrt{15^2 - 9^2} = \sqrt{144} = 12$. Option (B). (Recognize the 3-4-5 triangle times 3.)
Answer: (B)

56. Medium · No calculator needed

An angle has a measure of $\dfrac{7\pi}{12}$ radians. What is the measure of the angle, in degrees?

1. Multiply by $\frac{180^\circ}{\pi}$: $\frac{7\pi}{12} \cdot \frac{180}{\pi} = 7 \cdot 15 = 105$ degrees.
Answer: 105

57. Medium · No calculator needed

The number of radians in a 540-degree angle can be written as $a\pi$, where a is a constant. What is the value of a?

1. Multiply by $\frac{\pi}{180}$: $540 \cdot \frac{\pi}{180} = 3\pi$, so $a = 3$.
Answer: 3